



배관 차단장치의 구조적 안정성 검증 용역 < 900A >

연구책임자	서명
성명: 윤준용 소속: 한양대학교 공학대학 기계공학과 직위: 교수	
성명: 손영우 소속: 한국건설관리공사 산업기계설비기술사	

Part 1

- 연구 개요

Part 2

- 해석 개요

Part 3

- 해석 결과

Part 4

- 결론

◆ 연구 목표

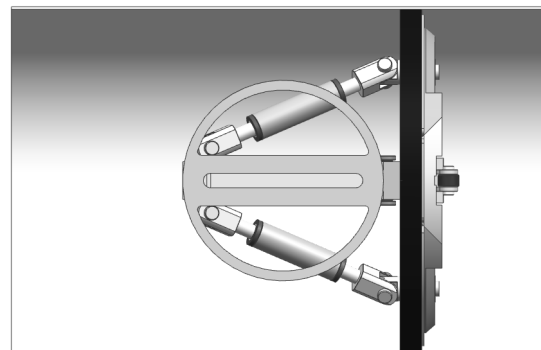
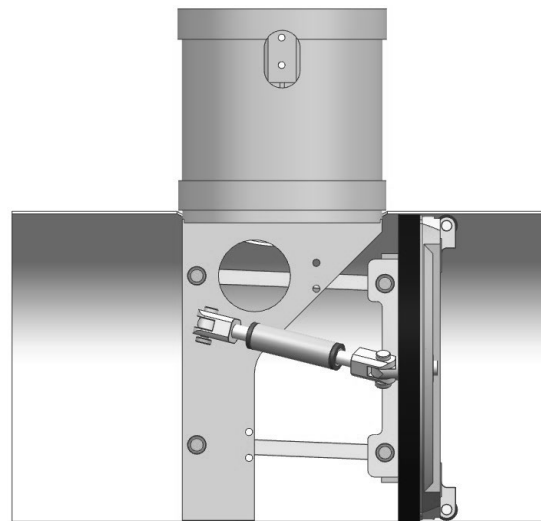
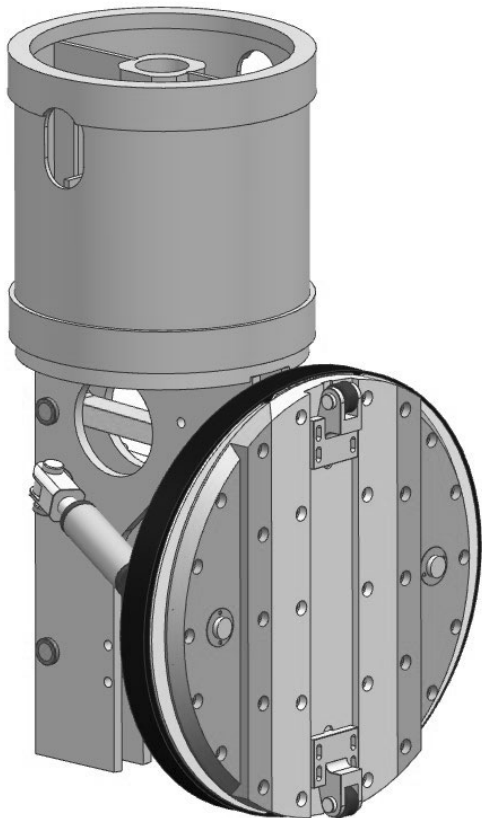
- 배관 차단장치의 구조적 안정성 평가

◆ 연구 수행 내용

- 배관 차단장치 3D 모델링
 - 구조해석 진행 전, 모델링 상용 프로그램인 ANSYS Spaceclaim을 사용하여 배관 차단장치를 3차원 형상으로 구현하는 작업
- 배관 차단장치 구조해석
 - 구조해석 상용 프로그램인 ANSYS Static Structural을 사용하여 하중에 따른 배관 차단장치의 변형 및 응력 계산
- ❖ 배관 차단장치 구조해석을 위한 형상 및 재료는 에서 제공한 도면(5%의 진원도)을 활용함.
- ❖ 본 해석은 각 요소(component)의 평균 응력을 기반으로 안정성 평가를 진행함.
- ❖ 본 해석의 평가는 서지압력(surge pressure) 또는 수격 현상(water hammer) 등의 이상 하중(unusual load)과 도면에 포함되지 않은 외적인 사항은 고려하지 않음.

◆ 해석 모델

- Model cases for Research
 - 900A



◆ 해석 조건

- Partial Structural Analysis in Steady State
 - coupled with other parts for numerical efficiency and accuracy

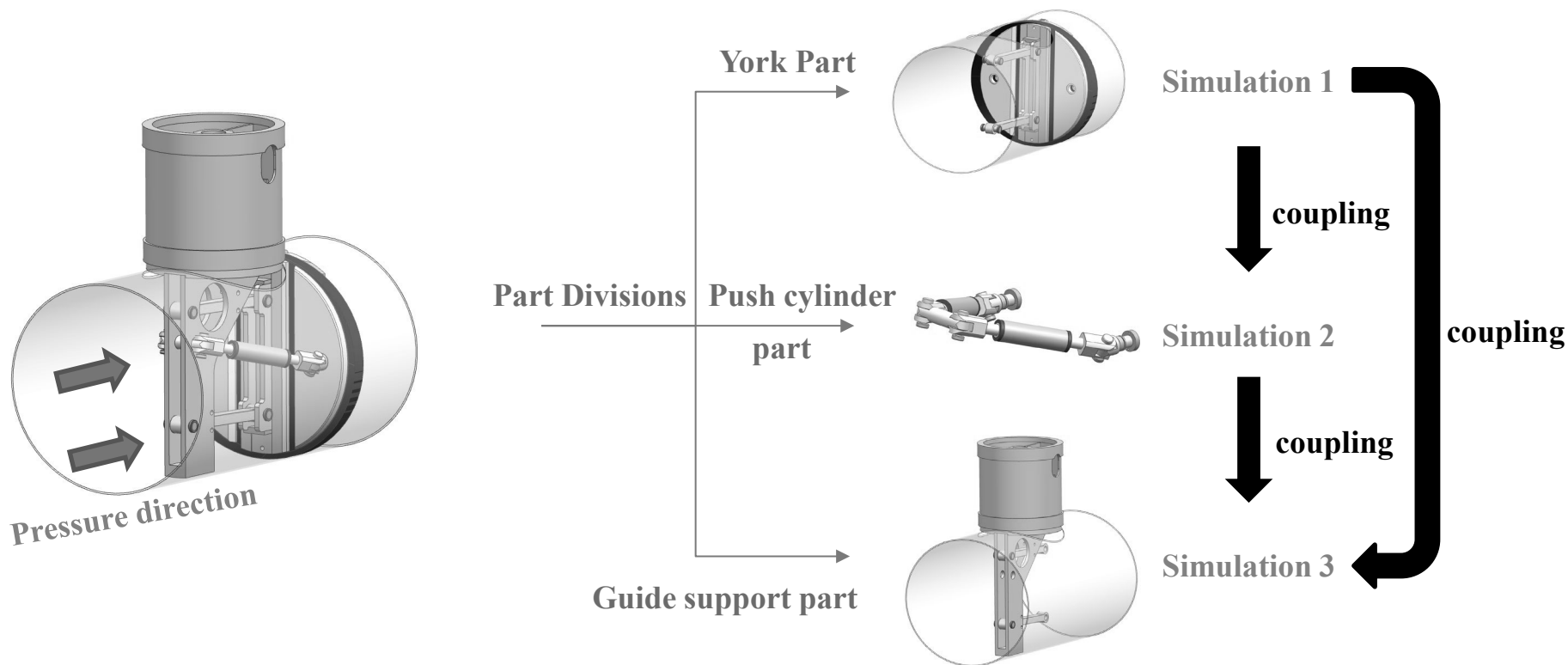


Fig. 1 Schematic diagram of numerical analysis procedure

◆ 해석 조건

- York part
 - Components

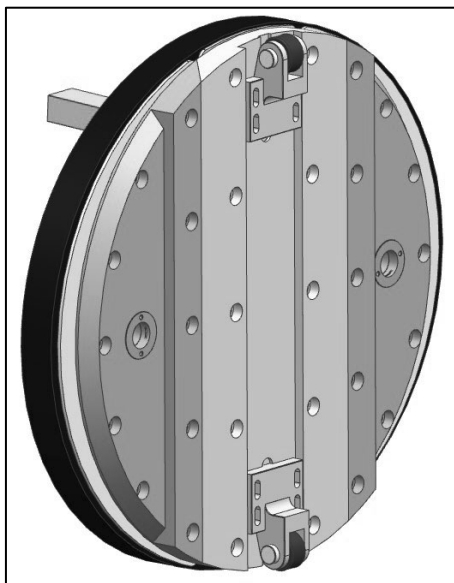


Fig. 2 York part

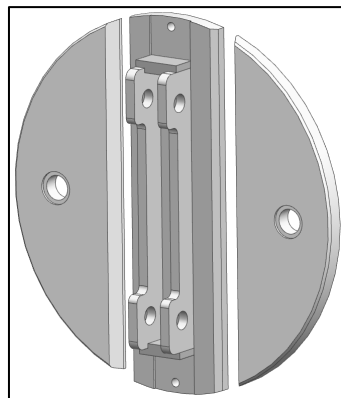


Fig. 3 Element plate

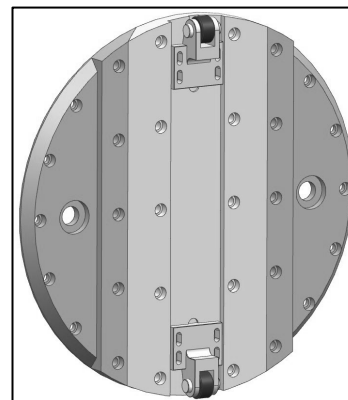


Fig. 4 York & Roller

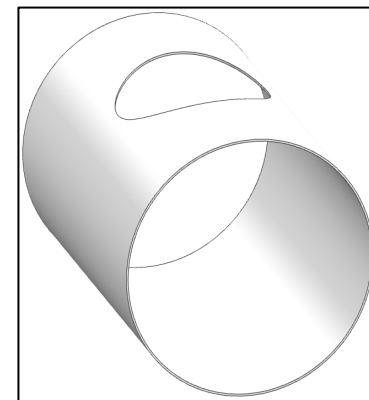


Fig. 5 Pipe

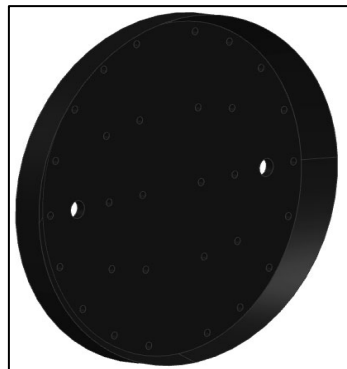


Fig. 6 Element

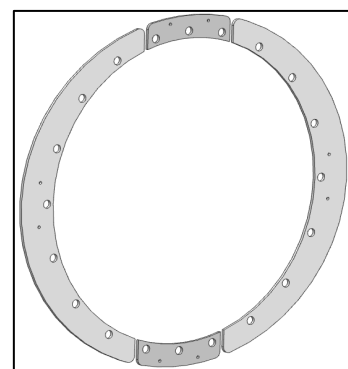


Fig. 7 York bracket

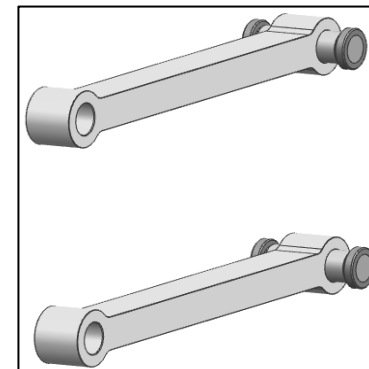


Fig. 8 Link & Link Pin

◆ 해석 조건

- York part
 - Material properties of York part components

Part	Material	Density (kg/m^3)	Young`s modulus(MPa)	Yield stress (MPa)	Tensile stress (MPa)	Poisson`s ratio
Element plate	A537 CL1	7820	210000	345	553	0.3
Link pin	S45C	7820	210000	490	686	0.3
Link	S45C	7820	210000	490	686	0.3
Yoke	A537 CL2	7820	210000	415	620	0.3
Element	RUBBER		70			0.49
York bracket	SS275	7820	210000	235	455	0.3

◆ 해석 조건

- York part
 - Boundary condition
 - ① pressure : $15\text{kgf/cm}^2 (\approx 1.47\text{MPa})$
 - ❖ Pressure direction is opposite to surface normal direction

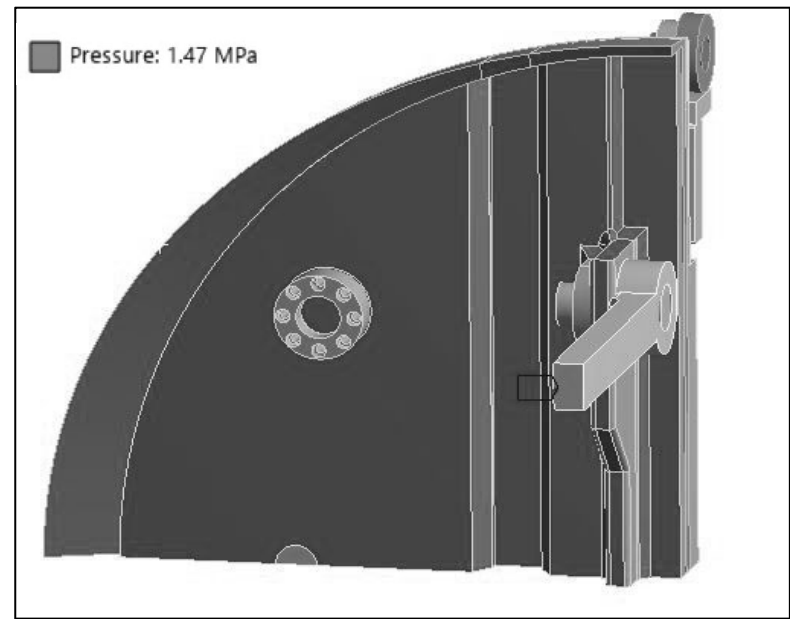
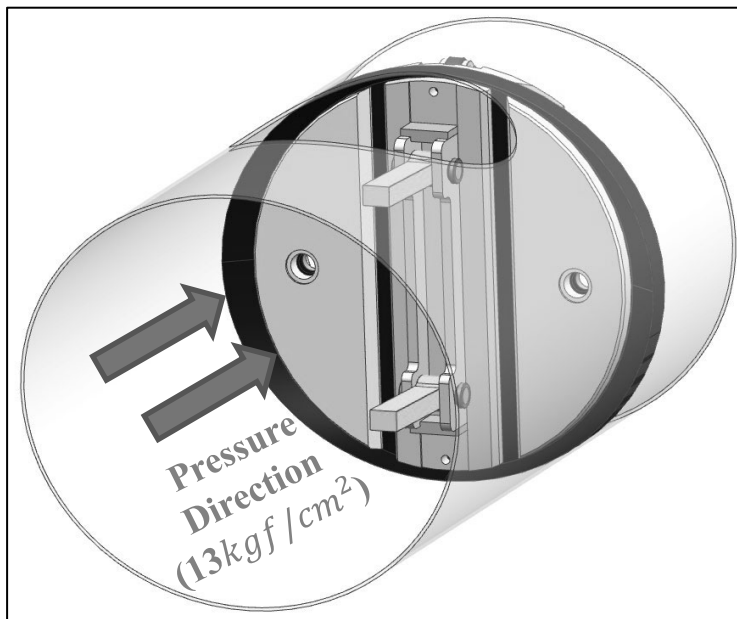


Fig. 9 Yoke part load condition

◆ 해석 조건

- York part
 - Constraints
 - ① Fixed support at the ends of Pipe surface for numerical singular problem
 - ② Frictional conditions between Pipe and Elements

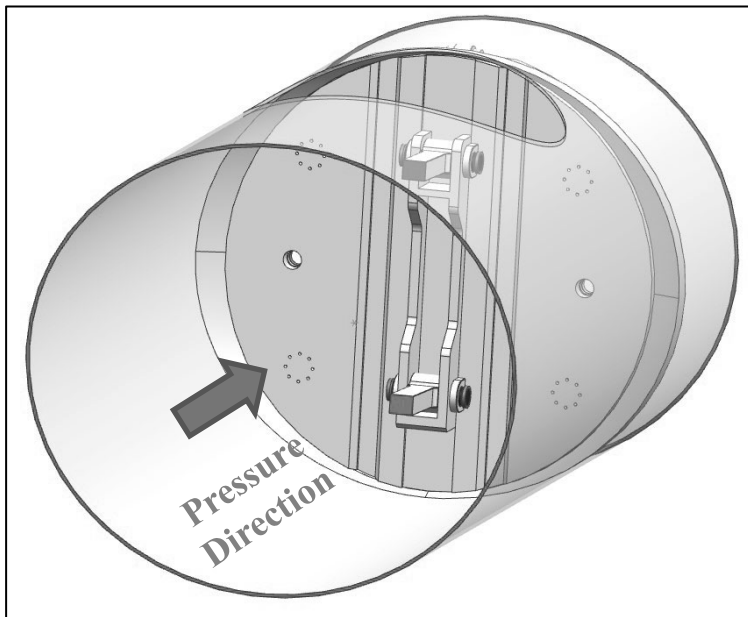


Fig. 10 constraint ① shown as red surfaces

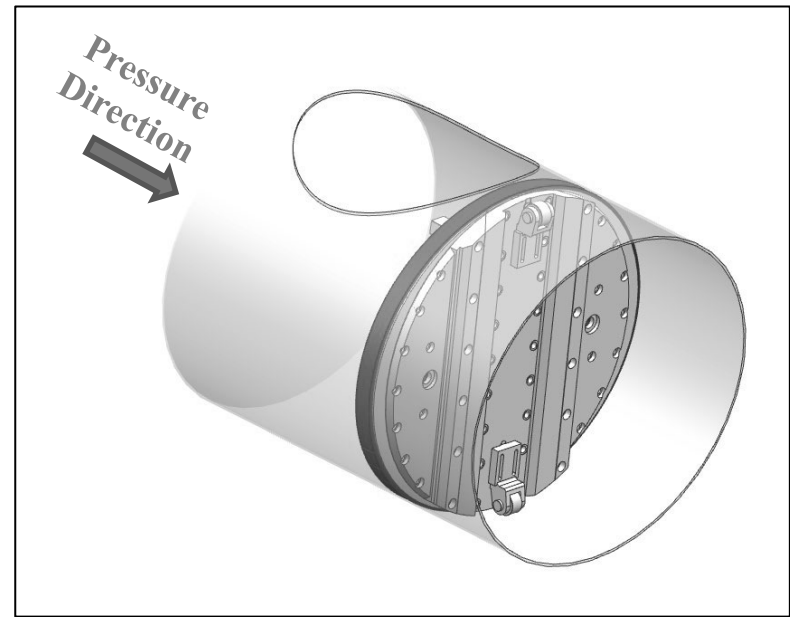


Fig. 11 constraint ② shown as red surfaces

◆ 해석 조건

- York part
 - Constraints
 - ③ Frictional conditions between each part surface
 - ④ Line bonded condition for combining each parts

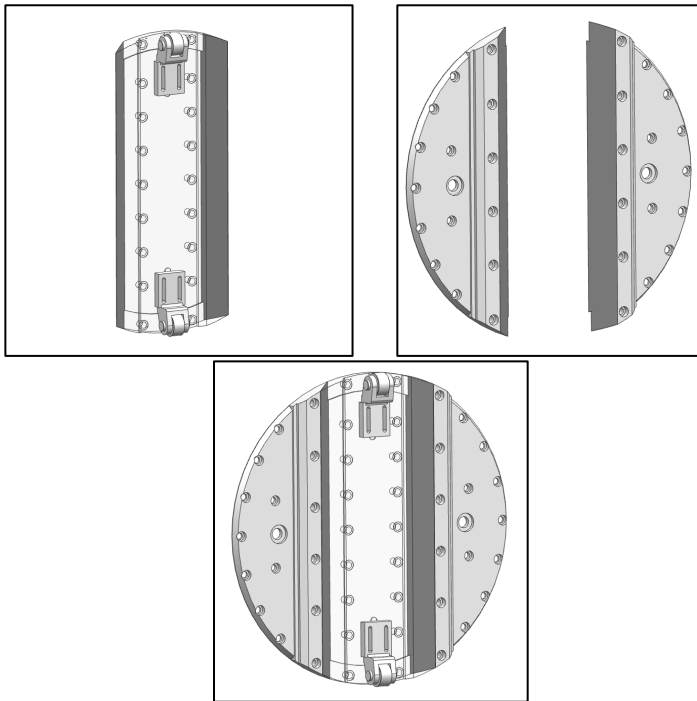


Fig. 12 constraint ③ shown as red and blue surfaces

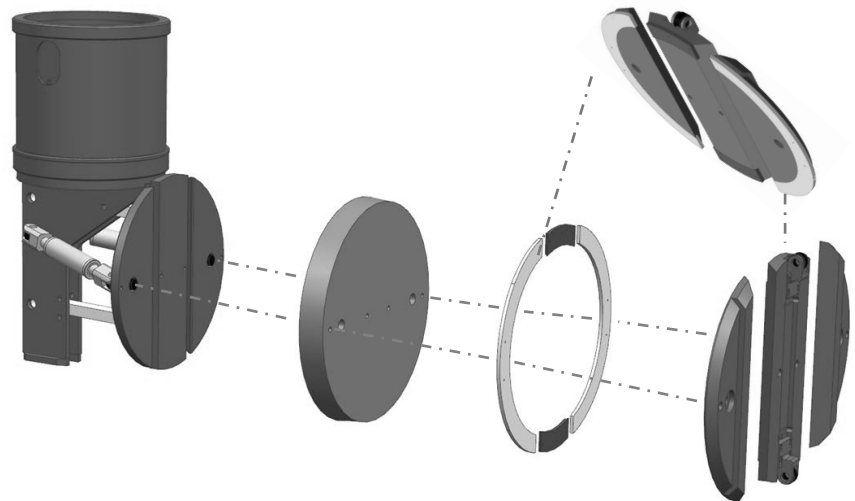


Fig. 13 constraint ④

◆ 해석 조건

- Push cylinder part
 - Components

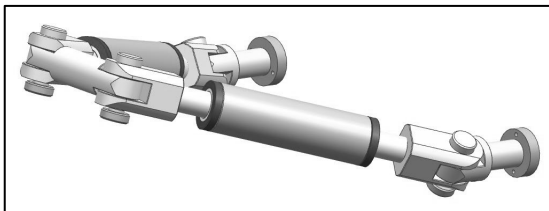


Fig. 14 Push part

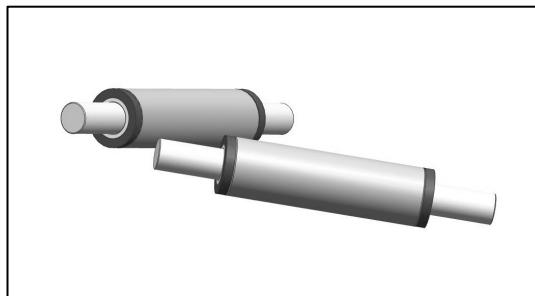


Fig. 15 Main push & Support Push

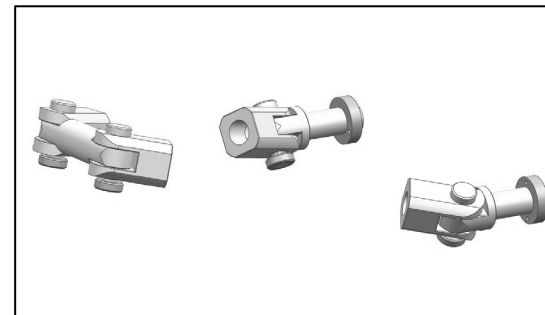


Fig. 16 Pin & Adapter & Cover

◆ 해석 조건

- Push cylinder part
 - Material properties of Push part components

Part	Material	Density (kg/m^3)	Young`s modulus(MPa)	Yield stress (MPa)	Tensile stress (MPa)	Poisson`s ratio
Push rod	S45C	7820	210000	490	686	0.3
Cover	S45C	7820	210000	490	686	0.3
Adapter	S45C	7820	210000	490	686	0.3
Pin	S45C	7820	210000	490	686	0.3

◆ 해석 조건

- Push cylinder part
 - Boundary condition
 - ① Displacement indicated as vectors obtained from York part results

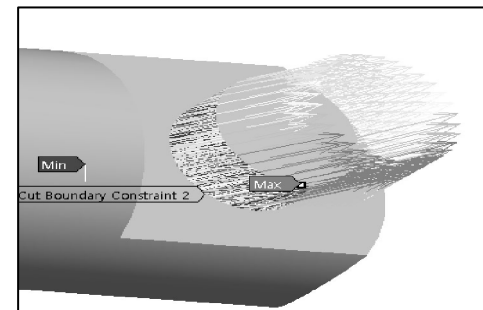
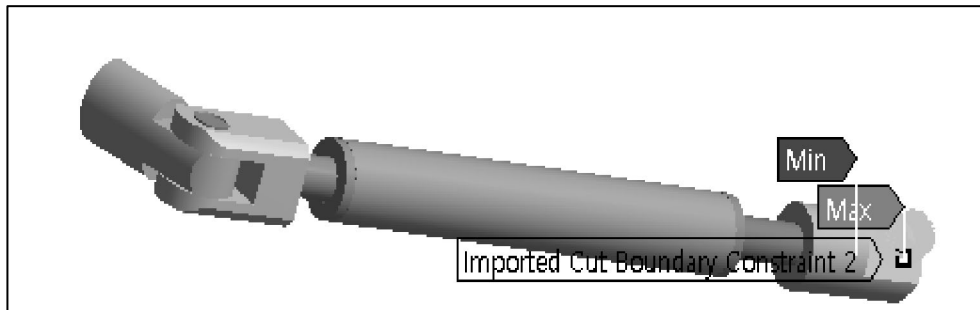


Fig. 17 Push part load condition

◆ 해석 조건

- Push cylinder part
 - Constraints
 - ① Allowance for revolution
 - ② Allowance for translation

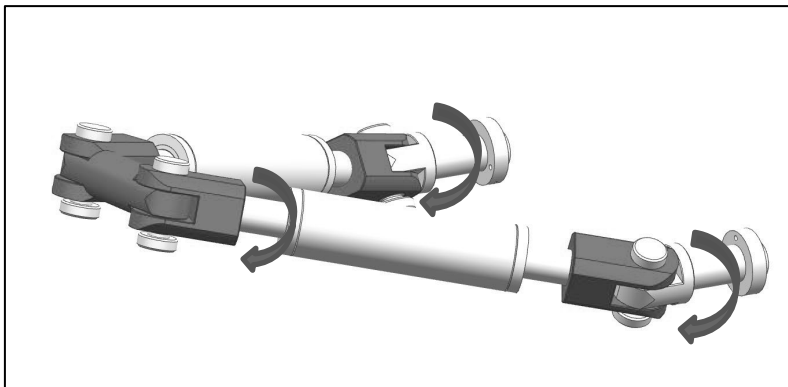


Fig. 18 constraint ① shown as red

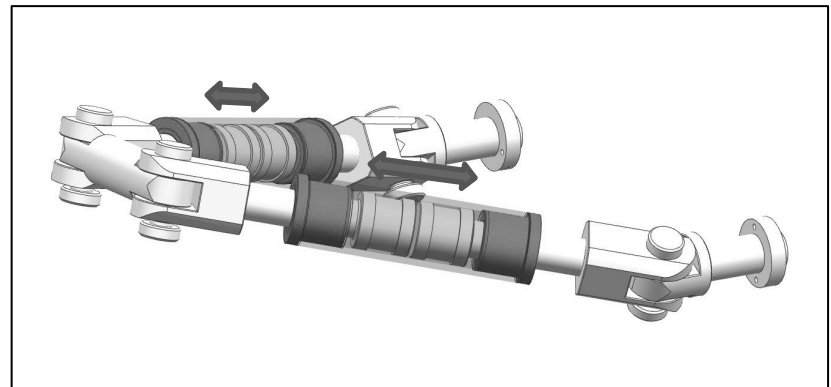


Fig. 19 constraint ② shown as red

◆ 해석 조건

- Guide support part
 - Components

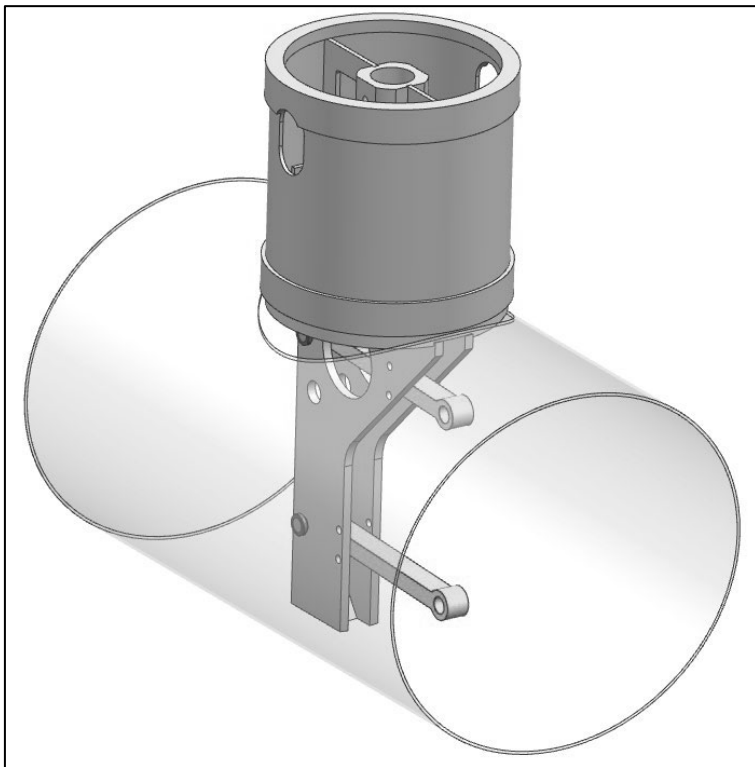


Fig. 20 Guide support part

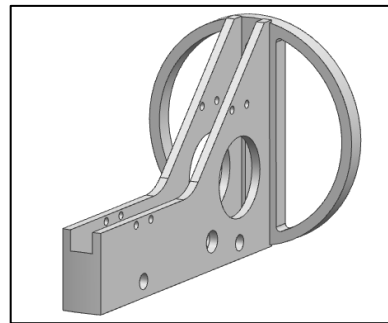


Fig. 21 Guide support

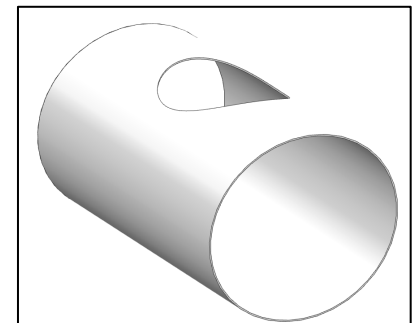


Fig. 23 Pipe

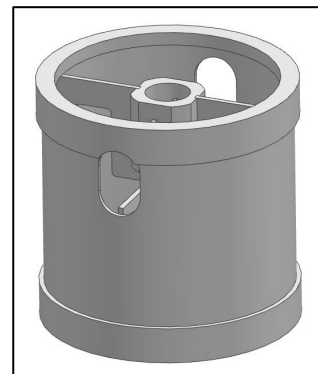


Fig. 22 Circular guide

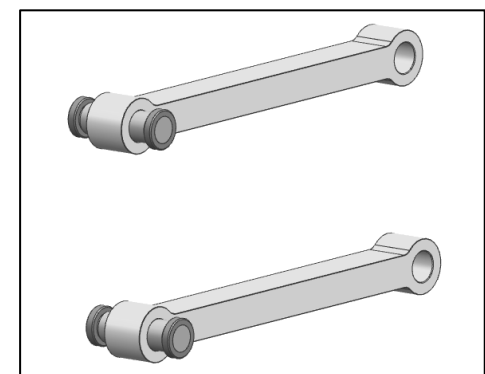


Fig. 24 Link & Pin

◆ 해석 조건

- Guide support part
 - Material properties of Guide support part components

Part	Material	Density (kg/m^3)	Young`s modulus (MPa)	Yield stress (MPa)	Tensile stress (MPa)	Poisson`s ratio
Guide support	SS275	7820	210000	275	480	0.3
Circular guide	SS275	7820	210000	275	480	0.3
Link	S45C	7820	210000	490	686	0.3
Pin	S45C	7820	210000	490	686	0.3

◆ 해석 조건

- Guide support part
 - Boundary condition
 - ① pressure : $15\text{kgf}/\text{cm}^2 (\approx 1.47\text{MPa})$
 - ❖ Pressure direction is opposite to surface normal direction

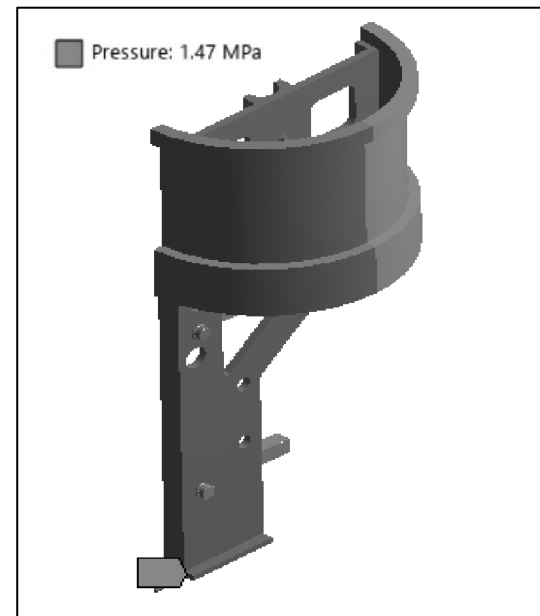
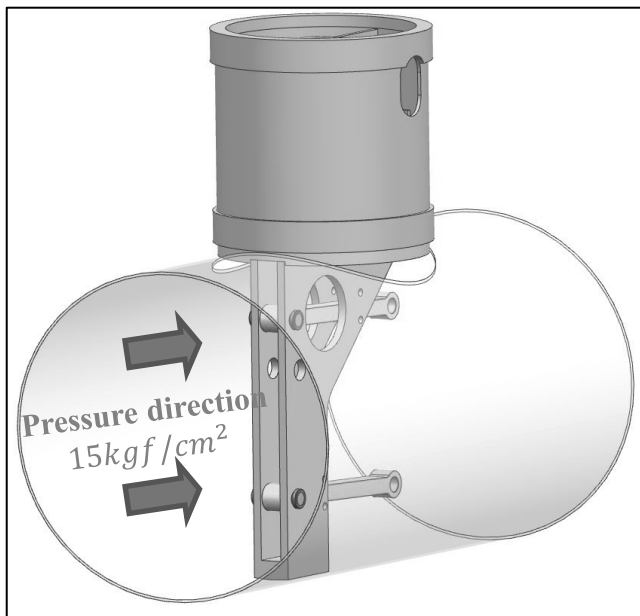


Fig. 25 Guide support part load condition

◆ 해석 조건

- Guide support part
 - Constraints
 - ① Fixed support at the ends of Pipe surface for numerical singular problem
 - ② Surface bonded condition for combining each parts

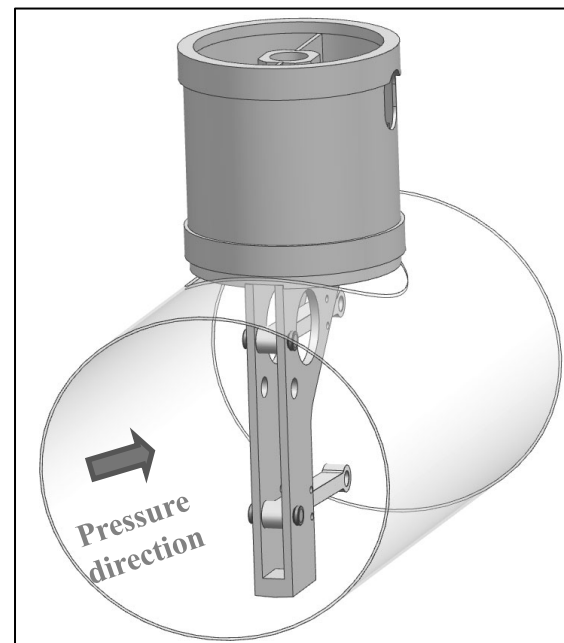
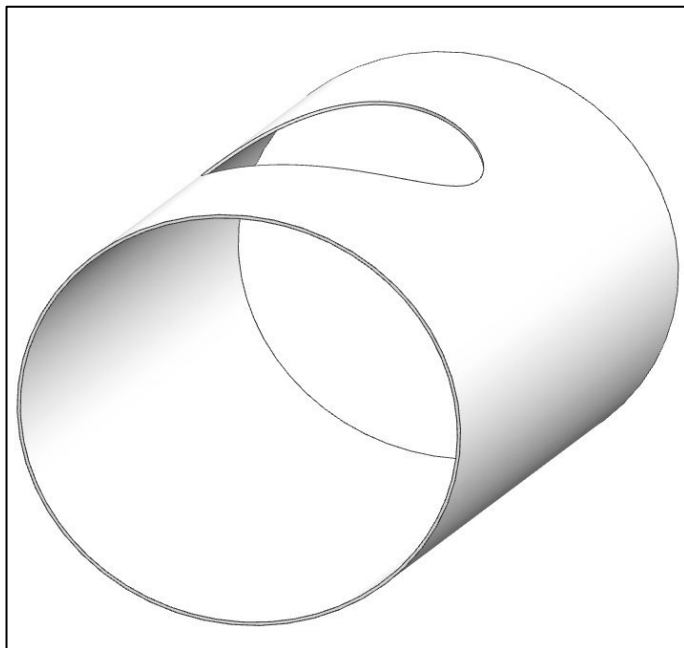


Fig. 26 constraint ① shown as red surfaces

◆ 해석 조건

- Methodology for Stress results

- Stress distribution, α

i : i th element of pipe surface

N : Total number of elements

σ_i : Stress on i th element(MPa)

A_i : Area of i th element(mm^2)

α_i : Stress distribution of i th element(%)

$$\alpha_i = \frac{\sigma_i A_i}{\sum_{k=1}^N \sigma_k A_k}$$

- Stress standard deviation, Z

i : i th element of a component

N : Total number of a component

σ_i : Stress on i th element(MPa)

σ_{avg} : Average stress on a component (MPa)

S : Variance of a component

Z_i : Z score of i th element

$$S = \sqrt{\frac{\sum_{i=1}^N (\sigma_i - \sigma_{avg})^2}{N}}$$

$$Z_i = \frac{\sigma_i - \sigma_{avg}}{S}$$

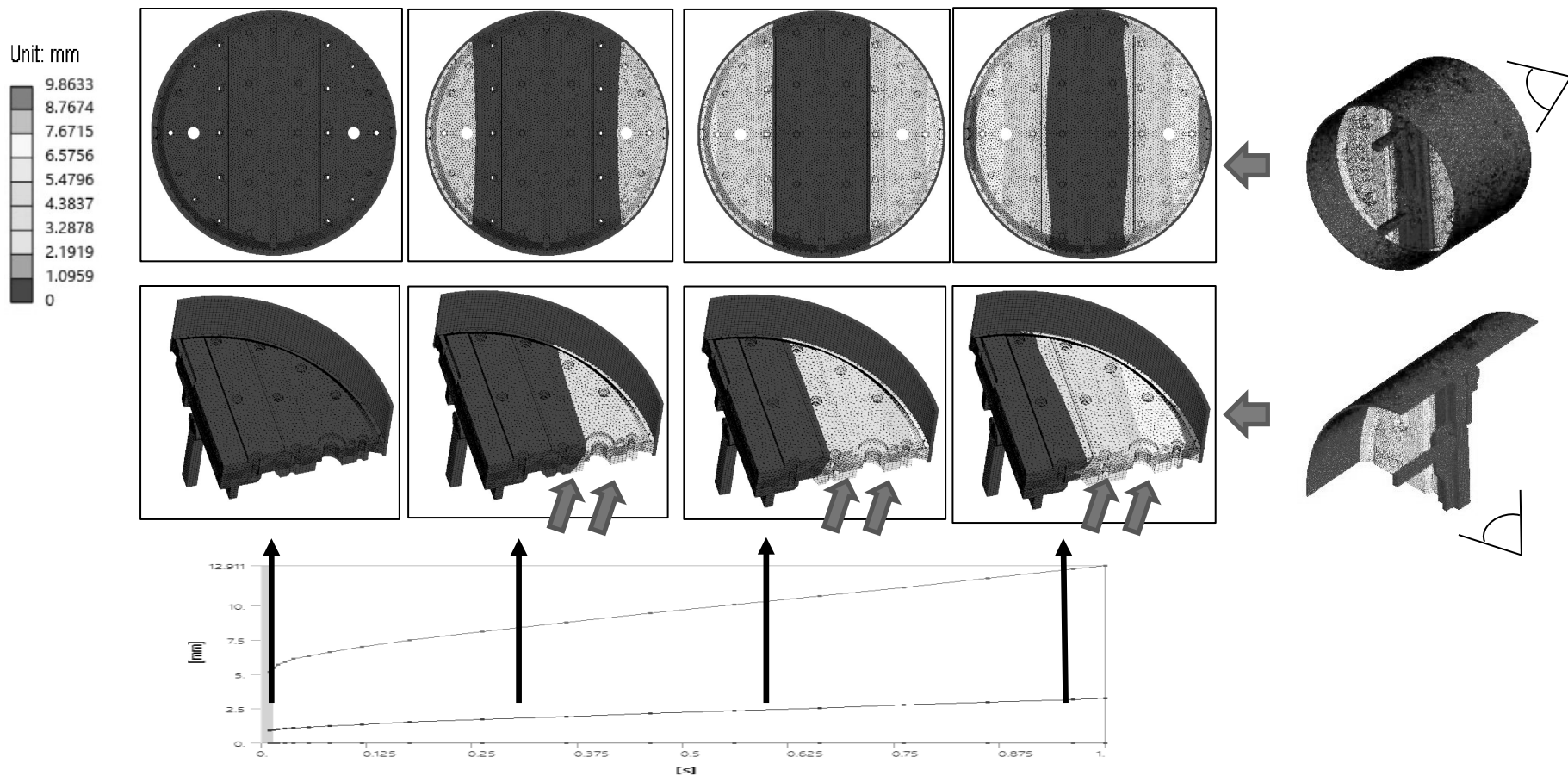
◆ 900A

- Yoke part

Part		Avg. Stress (MPa)	Max. Total Deformation (mm)
Element plate		31.09	9.86
Link and Pin		120.89	0.65
Element		1.26	9.73
Yoke		23.32	9.87
Yoke bracket		17.75	9.88

◆ 900A

- Yoke part
 - Total deformation : 9.86 mm



◆ 900A

- Yoke part
 - Element plate
 - ① Max. Total deformation : 9.86mm 12.10 mm
 - ② Avg. Stress : 31.09 MPa 21.84 MPa

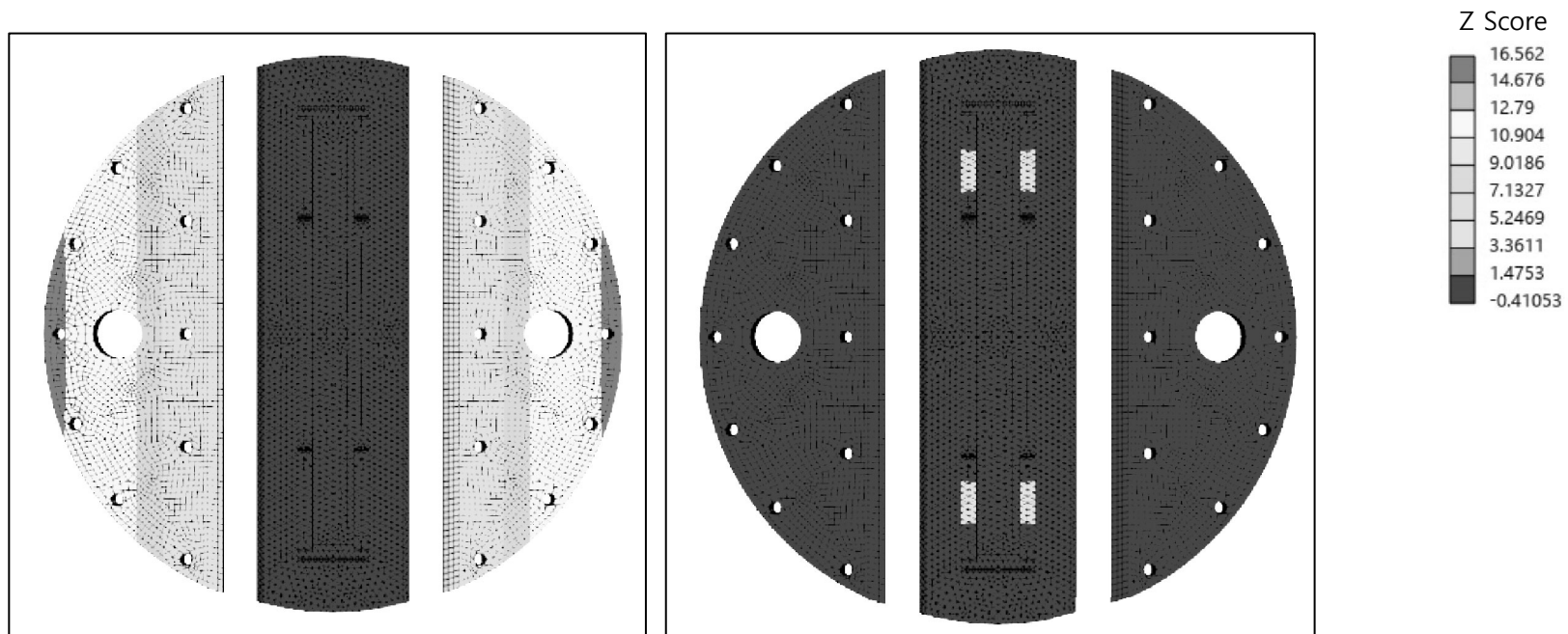
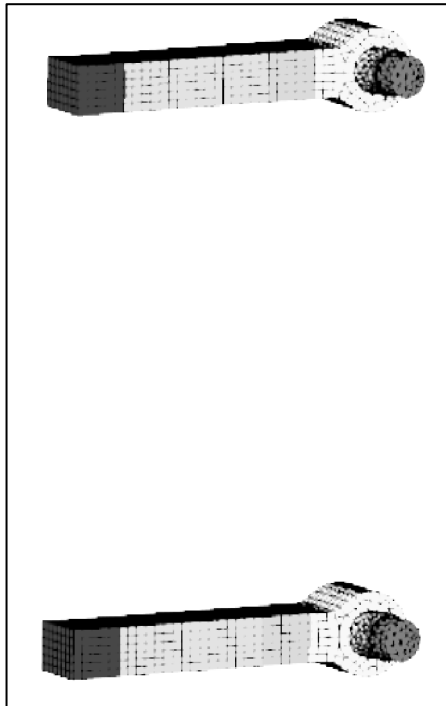
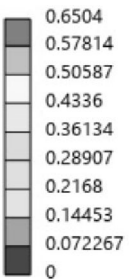


Fig. 29 Element plate results
(Left – Deformation / Right – Stress standard deviation)

◆ 900A

- Yoke part
 - Link & Pin
 - ① Max. Total deformation : 0.65mm 0.20591 mm
 - ② Avg. Stress : 120.89MPa 88.53 MPa

Unit: mm



Z Score

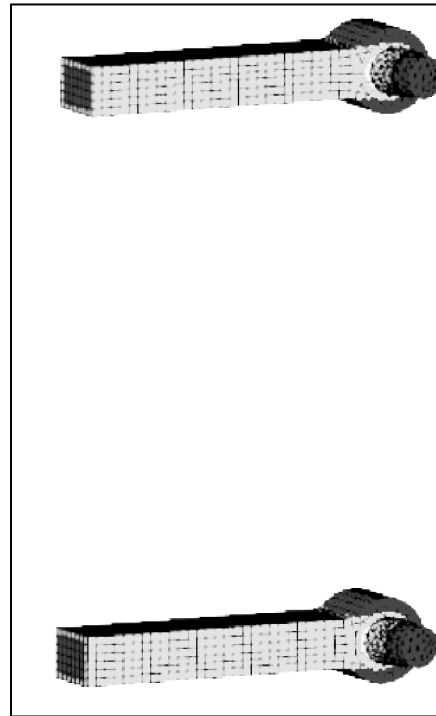
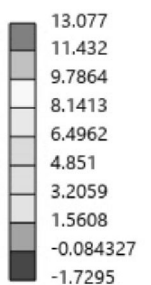
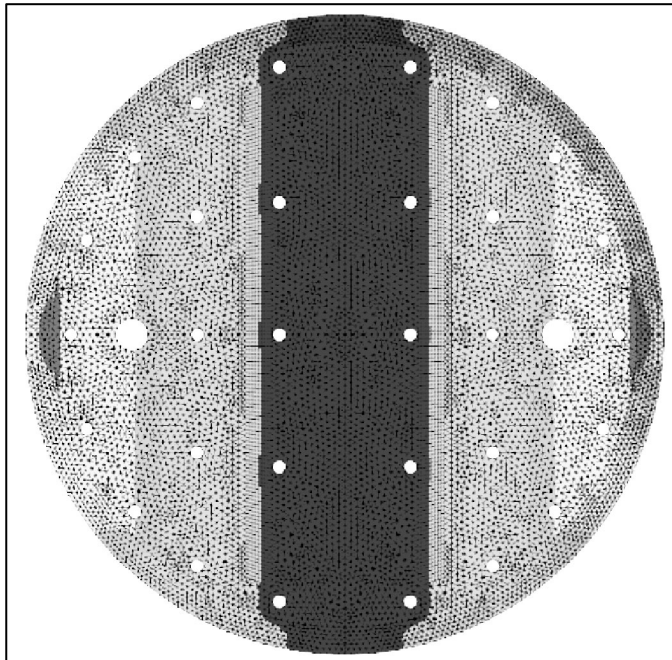
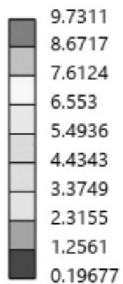


Fig. 30 Link & Pin results
(Left – Deformation / Right – Stress standard deviation)

◆ 900A

- Yoke part
 - Element
 - ① Max. Total deformation : 9.73mm 11.83 mm
 - ② Avg. Stress : 1.26MPa 1.98 MPa

Unit: mm



Z Score

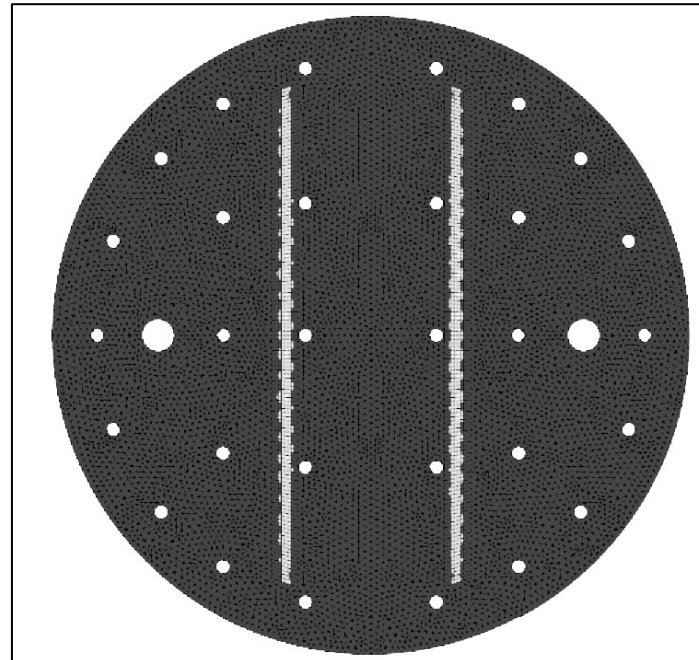
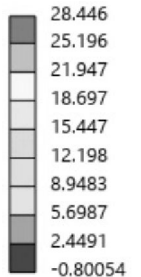


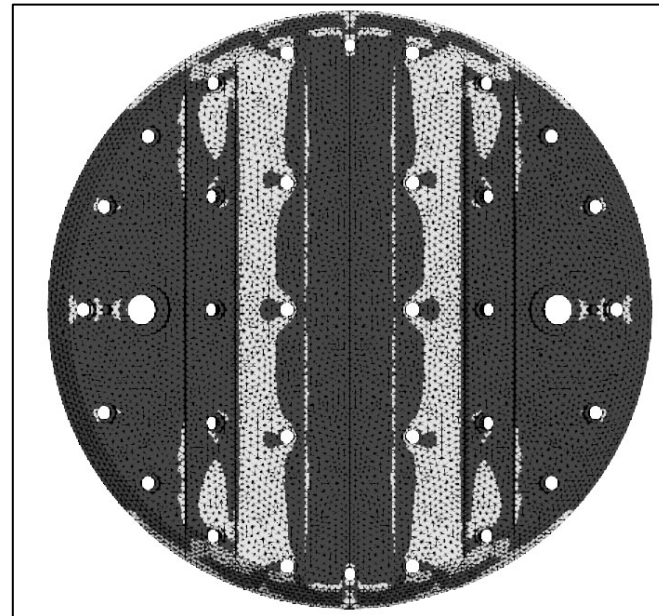
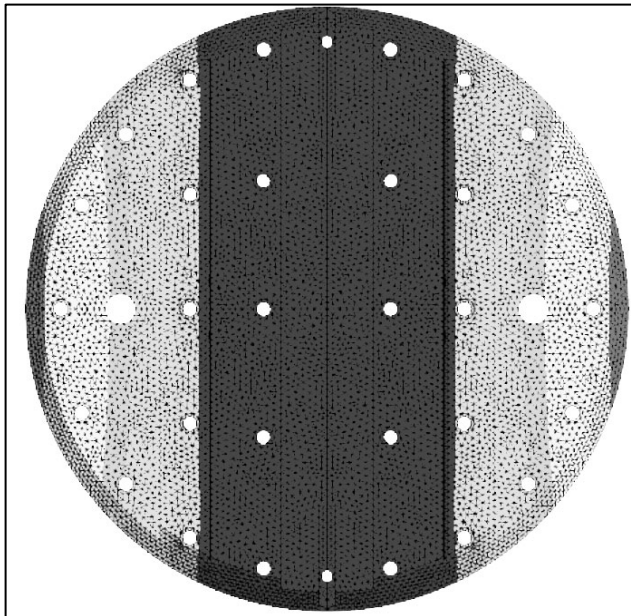
Fig. 31 Element results
(Left – Deformation / Right - Stress standard deviation)

◆ 900A

- Yoke part
 - Yoke
 - ① Max. Total deformation : 9.87mm 11.91mm
 - ② Avg. Stress : 23.32MPa 19.58 MPa

Unit: mm

9.8633
8.8468
7.8303
6.8138
5.7973
4.7808
3.7643
2.7478
1.7313
0.71482



Z Score

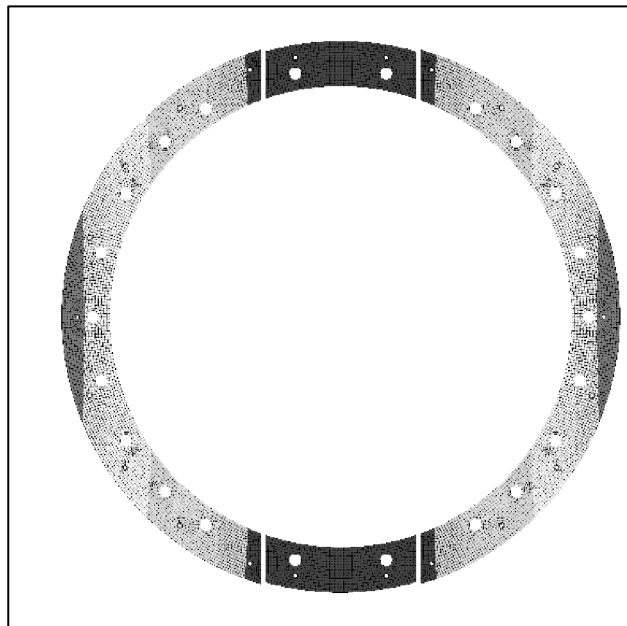
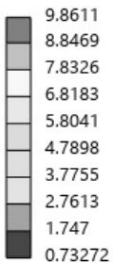
7.808
6.8456
5.8833
4.921
3.9586
2.9963
2.0339
1.0716
0.10926
-0.85308

Fig. 32 York bracket results
(Left – Deformation / Right - Stress standard deviation)

◆ 900A

- Yoke part
 - Yoke bracket
 - ① Max. Total deformation :9.88mm12.91 mm
 - ② Avg. Stress :17.75 MPa 13.88 MPa

Unit: mm



Z Score

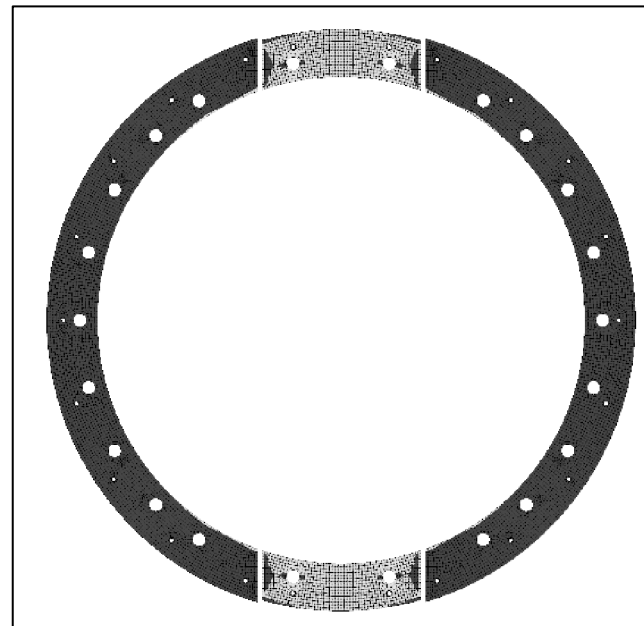
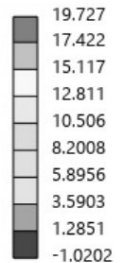


Fig. 33 York bracket results
(Left – Deformation / Right - Stress standard deviation)

◆ 900A

- Push cylinder part

Part		Avg. Stress (MPa)	Max. Total Deformation (mm)
Push rod		1.08	7.27
Pin & Adapter		2.20	7.06

◆ 900A

- Push cylinder part
 - Total deformation : 9.68 mm

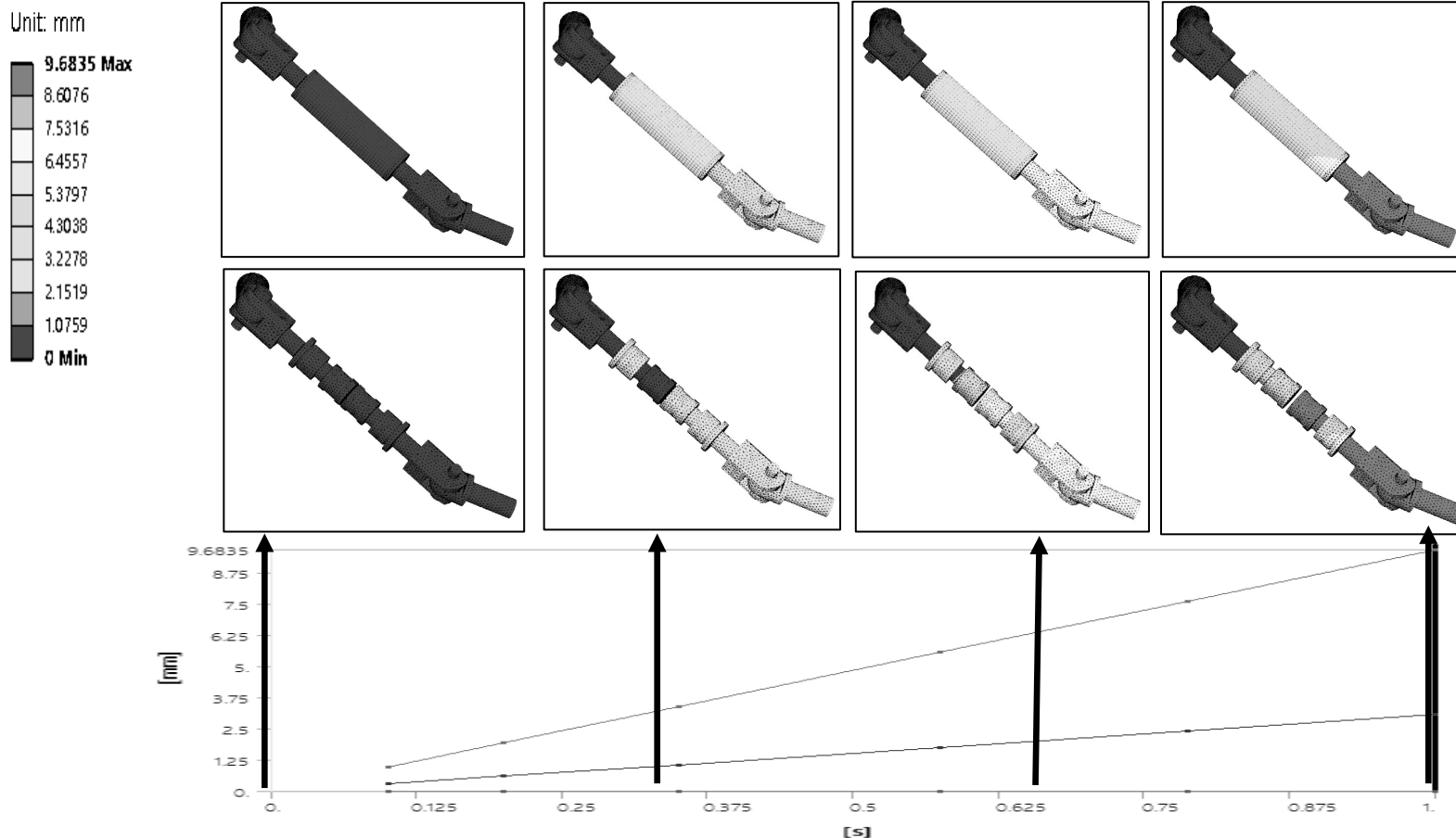


Fig. 34 Deformation results of Push part

◆ 900A

- Push cylinder part
 - Total deformation : 9.68 mm

Unit: mm

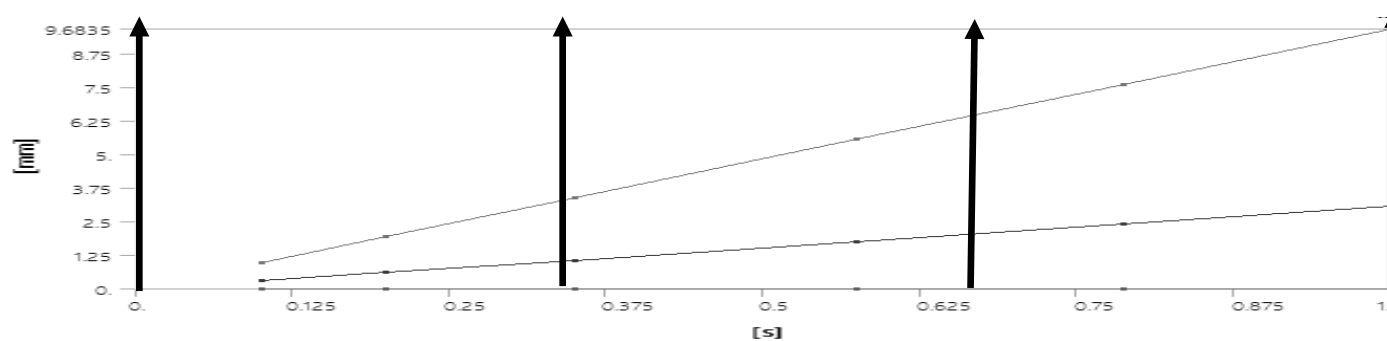
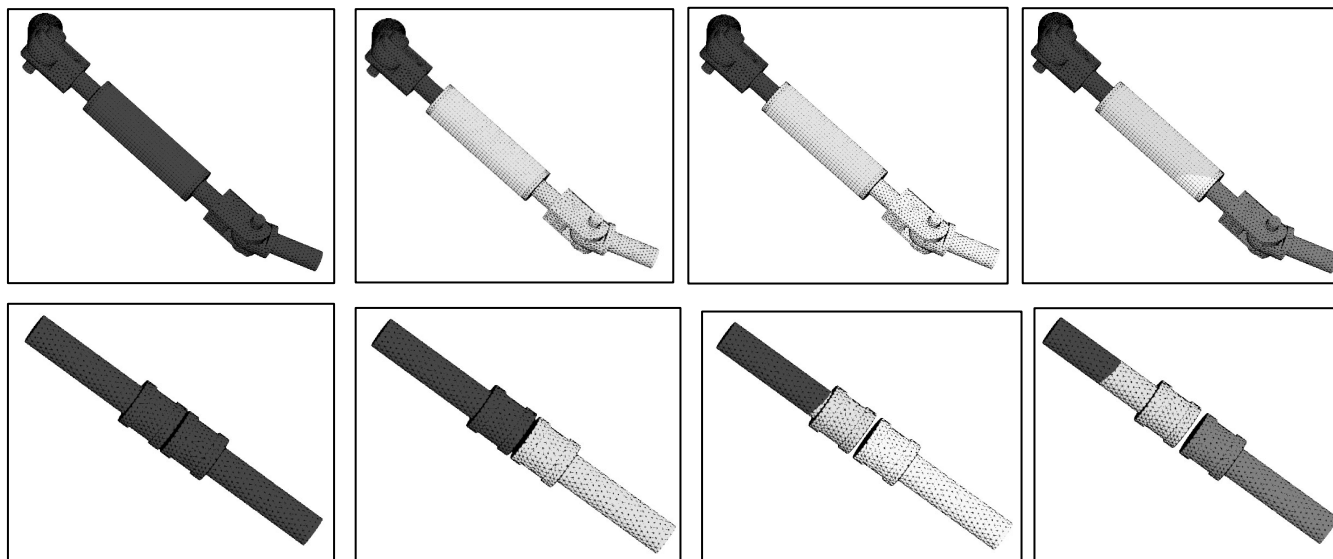
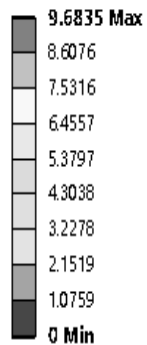


Fig. 34 Deformation results of Push part

◆ 900A

- Push cylinder part
 - Pin & Adapter
 - ① Max. Total deformation : 7.27mm 9.68 mm
 - ② Avg. Stress :1.08MPa 0.98 MPa

Unit: mm

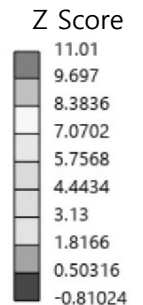
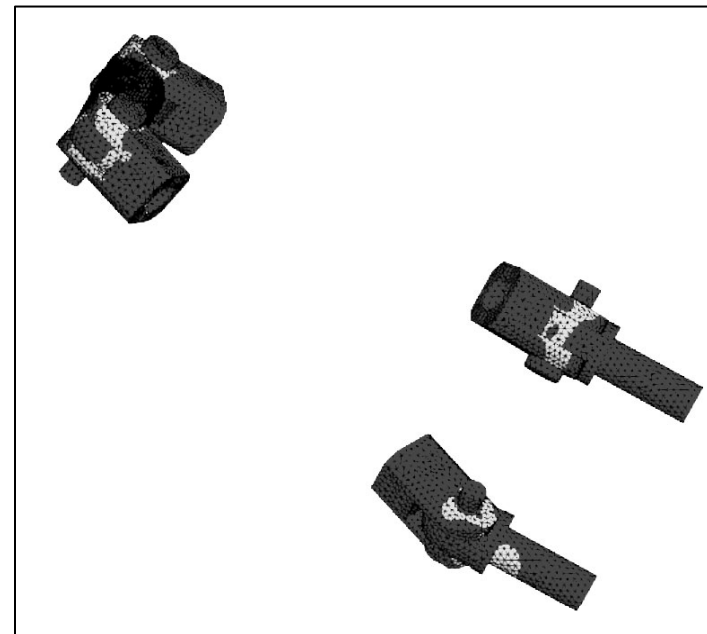
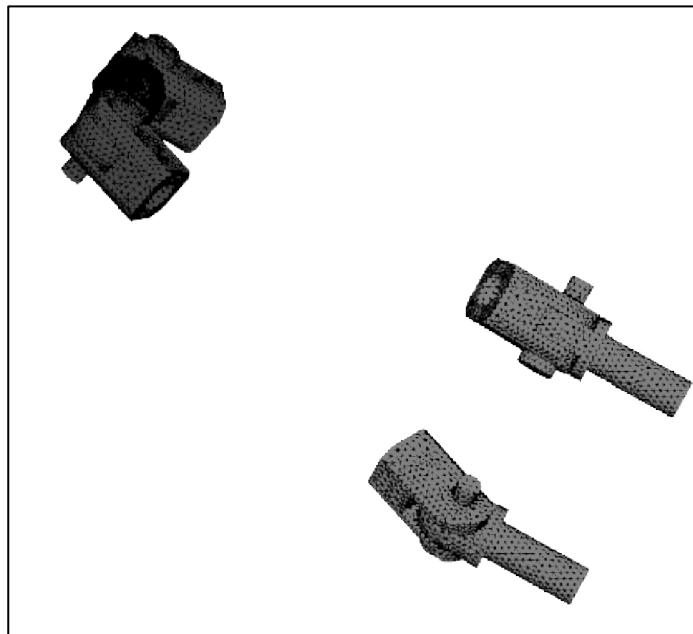
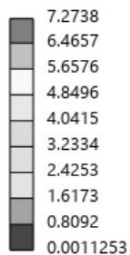


Fig. 35 Pin & Adapter results
(Left – Deformation / Right - Stress standard deviation)

◆ 900A

- Push cylinder part
 - Push rods
 - ① Max. Total deformation :7.06mm 9.68 mm
 - ② Avg. Stress :1.16MPa 2.20 MPa

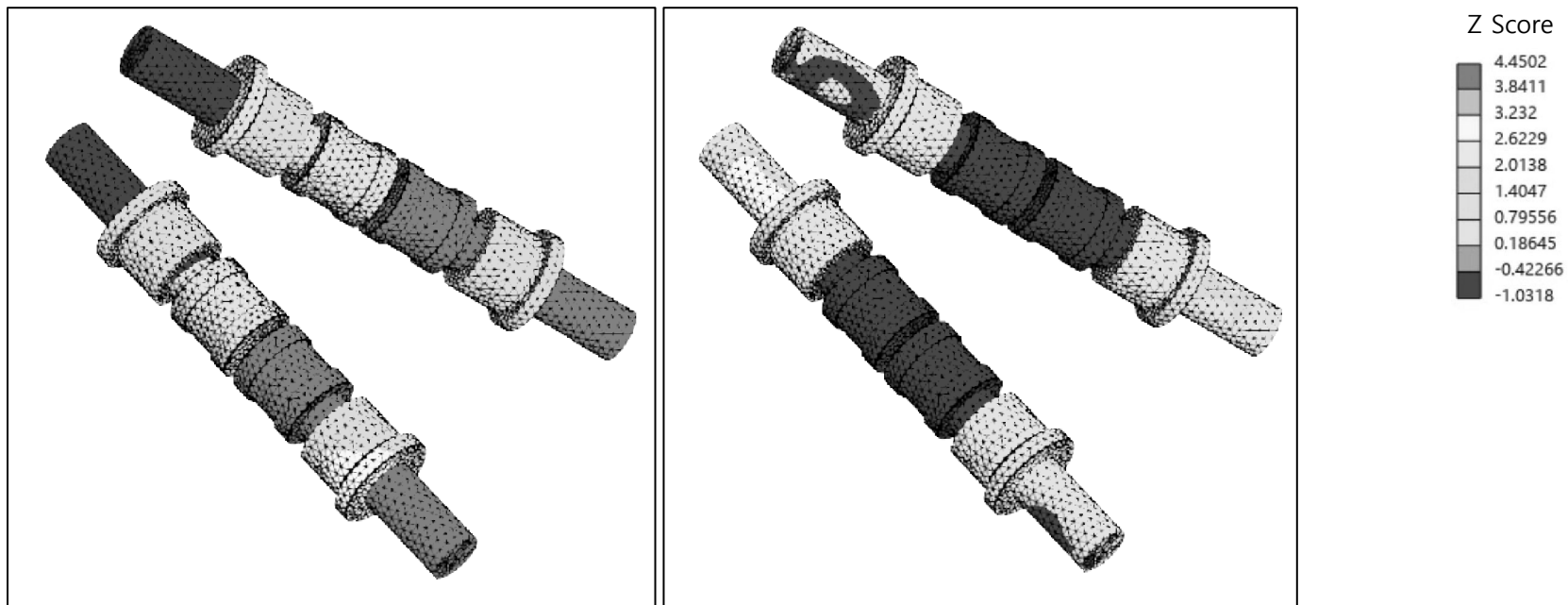
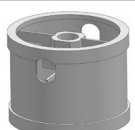


Fig. 36 Push rods results
(Left – Deformation / Right - Stress standard deviation)

◆ 900A

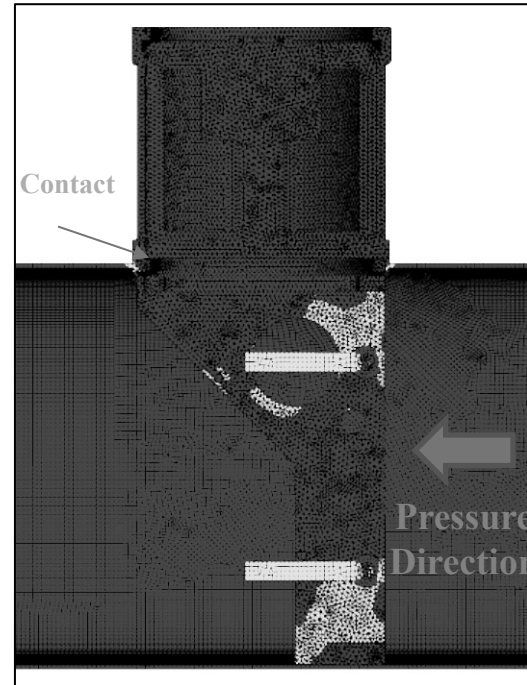
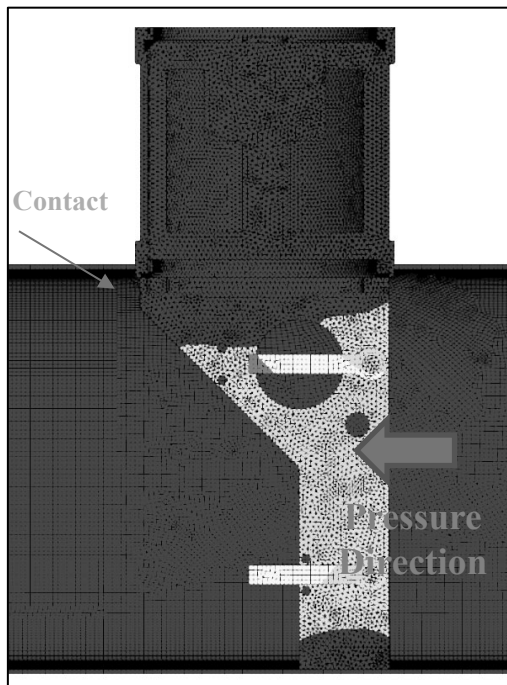
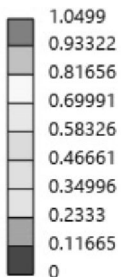
- Guide support part

Part		Avg. Stress (MPa)	Max. Total Deformation (mm)
Guide support		34.02	0.56
Circular guide		2.94	0.79
Link & Pin		49.34	1.05

◆ 900A

- Guide support part
 - Overall component
 - ① Max. Total deformation : 1.05mm 1.56 mm
 - ② Avg. Stress : 22.58MPa 11.66 MPa

Unit: mm



Z Score

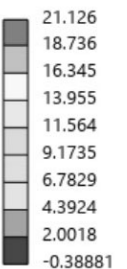


Fig. 37 Overall components results
(Left – Deformation / Right - Stress standard deviation)

◆ 900A

- Guide support part
 - Circular guide
 - ① Max. Total deformation :0.79mm 1.56 mm
 - ② Avg. Stress :2.94MPa 2.82 MPa

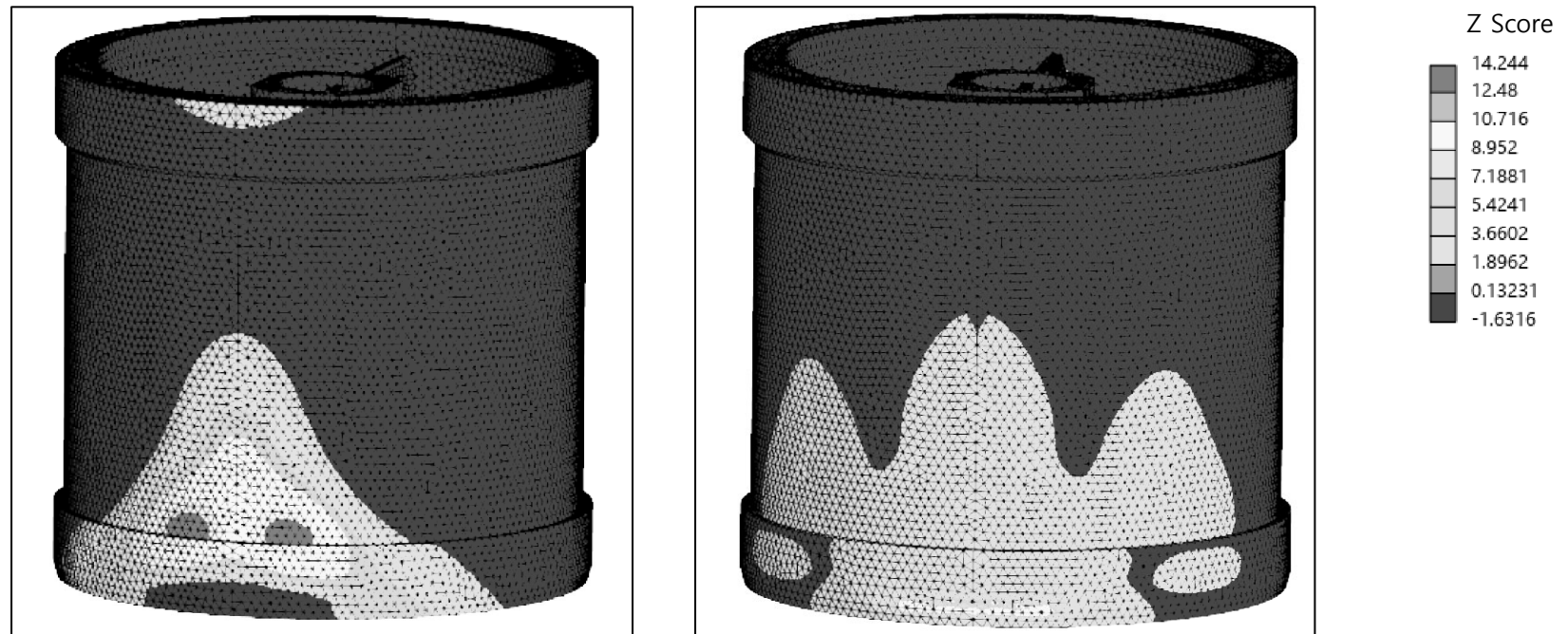
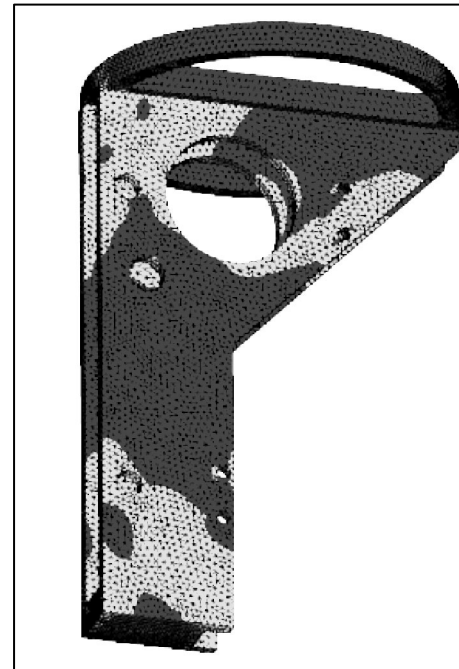
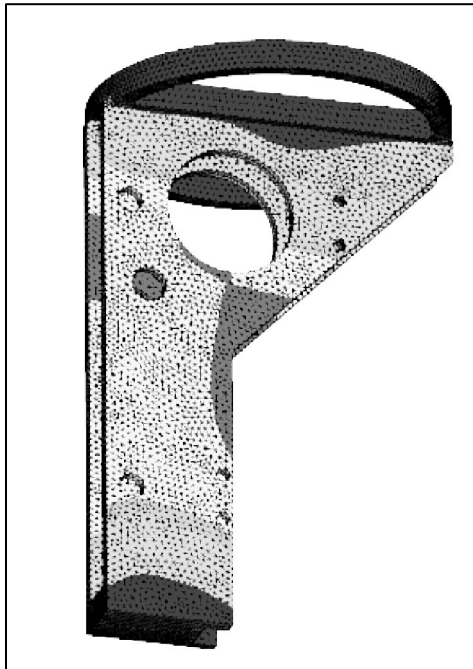
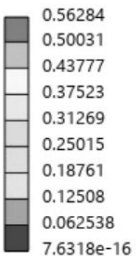


Fig. 38 Circular guide results
(Left – Deformation / Right - Stress standard deviation)

◆ 900A

- Guide support part
 - Guide support
 - ① Max. Total deformation : 0.56mm 1.39 mm
 - ② Avg. Stress : 34.02MPa 15.08 MPa

Unit: mm



Z Score

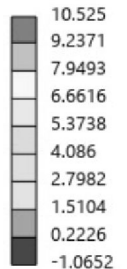
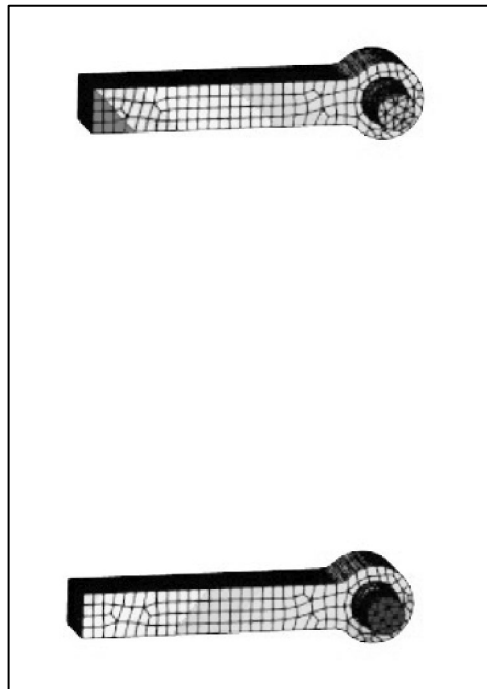
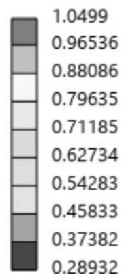


Fig. 39 Guide support results
(Left – Deformation / Right - Stress standard deviation)

◆ 900A

- Guide support part
 - Link and Pin
 - ① Max. Total deformation : 1.05mm 1.30 mm
 - ② Avg. Stress : 49.34MPa 32.12 MPa

Unit: mm



Z Score

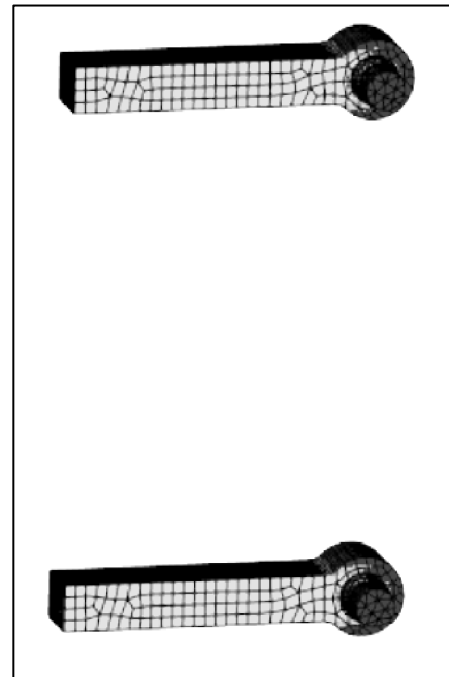
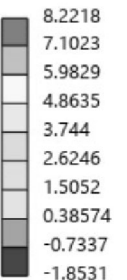


Fig. 40 Link and Pin results
(Left – Deformation / Right - Stress standard deviation)

◆ 구조해석을 통한 900A 배관 차단장치의 안정성 분석

• 구조해석 결과, 안정성 평가에 대한 결론은 다음과 같다.

- ① 각 요소의 평균 응력은 재료의 항복강도 이하로 계산됨.
- ② Yoke part의 경우, $15kgf/cm^2$ 압력 조건에서 유체 차단이 가능함을 확인함.
 - Yoke part가 이상적인 범위 내에서 변형되고 응력 분포가 균일하다
 - Rubber와 배관 내벽은 장치 작동 중에 밀착되어 효과적인 밀봉을 보장함
- ③ Push cylinder part의 경우, $15kgf/cm^2$ 압력 조건에서도 Push cylinder 사이의 간극이 존재하는 것을 확인함.
 - Push cylinder 내부의 모든 부품에 간섭이나 과도한 변형이 없음
 - 안전 여유가 높고 반복 운전 시 안정성 확보함
- ④ Guide support part의 경우, $15kgf/cm^2$ 압력 조건에서 Yoke part와 Push cylinder part를 안정적으로 지지하여 배관 차단장치의 정상적인 운전을 확보함
 - Guide support part내부의 각 부품에 경사와 오정렬이 나타나지 않아 효과적인 지지를 제공함